

Optimization of the Process Parameters for the Synthesis Process of Battery-Grade Ferrous Oxalate by Response Surface Method

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This study investigates the optimal conditions for the synthesis of battery-grade ferrous oxalate as a raw material for preparing cathode material. Ferrous oxalate was prepared by liquid-phase precipitation method using ferrous sulfate and oxalic acid. Central composite design (CCD) was used to determine the effects of three preparation variables on purity and particle size: reaction temperature, aging time and concentration of ferrous sulfate. Based on CCD, the significant factors on each experimental design response identified the analysis of variance (ANOVA). The optimum ferrous oxalate preparation conditions were obtained reaction temperature of 31.32°C, aging time of 56.52 min, and ferrous sulfate concentration of 5%. Under these optimum conditions, ferrous oxalate with purity of 99.69% and particle size of 4.92 μm was obtained as best product which met and exceed the requirements of battery-grade ferrous oxalate. In addition, the special morphologies of ferrous oxalate prepared under different dispersant proportion was characterized by scanning electron microscope (SEM) to analyze the mechanism of synthesis. Morphology control study revealed that the dispersant could effectively change the surface energy between crystallographic planes, then result in anisotropic growth of the crystal structure and change the morphology of synthetic products.

Keywords: Ferrous oxalate; response surface methodology (RSM); liquid-phase precipitation method; lithium ion battery; raw material.

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1. Introduction

Ferrous oxalate, a kind of chemical material, is widely used as chemical precursors of synthetic nanomaterials, ceramic superconductors or catalysts.¹ In recent years, with the fast development of lithium ion battery, ferrous oxalate, as a raw material for preparing cathode material of LiFePO_4 , also draws people's attention.^{2–5} On the other hand, ferrous oxalate has also been used for the coordination polymers (CPs) with high proton conductivity⁶ and a new anode material of lithium ion battery with a reversible capacity close to $700 \text{ mAh}\cdot\text{g}^{-1}$.^{7,8}

As the raw material of LiFePO_4 and the new anode material, the purity and particle size of ferrous oxalate affect the performance of lithium ion battery greatly.^{9,10} In a view of these facts, ferrous oxalate is prepared by liquid-phase precipitation method which is a simple and common pathway to synthesize magnetic nanoparticles.¹¹ Meanwhile, it is prominent that the nanoparticles are easy to aggregate due to rapid growth of crystal nucleus of liquid-phase precipitation method.¹² And, the preparation of ferrous oxalate was affected by numerous factors in this method. In previous studies, the effects of single factor such as aging temperature, ways of material feeding, pH value, dispersant amount on microstructure, particle size, and morphology of ferrous oxalate were considered.¹³ And as a new anode material of lithium ion battery, people concentrate on how to raise its electrochemical performance by using different surfactant to prepare ferrous oxalate with distinct morphology.¹⁴ However, there is hardly any studies on the optimization of preparation process and consider the interaction between multiple factors. In recent years, response surface methodology (RSM) is a powerful technique that explores the relationships and interactions among multiple variables.^{15,16} It is a collection of statistical and mathematical technique for developing, improving, and optimizing process. Finally, a group of regression equations with high precision are obtained to describe the relationship between responses and parameters by a series of experiments. This enables the delineation of an optimal response and characterization of experimental parameters.¹⁷ So, RSM is proved to be a powerful statistical tool to explore the optimization of multivariate experimental design.

In this study, significant factors were obtained and the optimum process parameters were investigated by RSM–central composite design (CCD). The effects of reaction temperature, aging time and concentration of ferrous sulfate were simultaneously studied to prepare ferrous oxalate with high purity and small particle size. In addition, the structural and morphological characterization of ferrous oxalate prepared under optimum reaction condition was also considered. It is expected to provide a well understanding for controlling the preparation process of ferrous oxalate.

2. Experiment

2.1. Materials

Analytical reagent (AR) iron (II) sulfate heptahydrate (99–101%) with content of Mn 0.037% and molecular weight (MW) 278.02 Da, and AR oxalic acid dihydrate ($\geq 99.5\%$) were used to prepare ferrous oxalate. Ethanol absolute ($\geq 99.7\%$) was used as the dispersant. Potassium permanganate standard solution with accurate concentration was prepared for titration analysis of purity. Deionized water was used for dissolving iron (II) sulfate heptahydrate and oxalic acid dihydrate, rinsing precipitate and analyzing product.

2.2. Experimental procedure

2.2.1. Ferrous oxalate preparation

Ferrous oxalate was prepared by using liquid-phase precipitation method. Iron (II) sulfate heptahydrate was dissolved deionized water. Oxalic acid dihydrate was dissolved by deionized water and dispersant with a certain proportion. In order to control the reaction time, oxalic acid solution was dropped from separatory funnel. Reaction temperature, stirring speed, reaction pH, and aging time were also controlled in this process. At the final stage, the precipitate of ferrous oxalate was separated by vacuum filtration device, then dried in oven at a suitable temperature and time, and finally the product would be obtained.

2.2.2. Single-factors experimental

Single factor test was used to examine the effect of each factor on preparation and determine the

preliminary range of reaction variables including reaction temperature, aging time, ferrous sulfate concentration, and dispersant proportion.¹⁸

Before preparing ferrous oxalate, ferrous sulfate were dissolved at the concentration of 5%, 10%, 15%, 20% and 25%, respectively. The dispersing agents were added into oxalic acid solutions at the proportion (the ratio of volume to water) from 20% to 100%. In preparation process, reaction temperatures were controlled at 10, 20, 30, 40, 50 and 60°C, respectively. After that, the aging time was considered to control from 20 min to 100 min. Finally, the samples were washed thoroughly with deionized water, and dried at 60°C for 8 h.

2.2.3. Response surface optimization and continuous mode treatment

Preparation process for ferrous oxalate was optimized through rigorous experimentation using RSM of Design Expert software. The optimum concentrations for reagent ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) and the optimum reaction conditions (aging time and reaction temperature) were determined. RSM is a statistical method that uses experimental data generated from specific experimental design to model and optimizes any process in which several variables influence the desired response.¹⁹ CCD was employed to design the experiments, this method helps to optimize the effective parameters with a minimum number of experiments and also analyze the interaction between the parameters and results.²⁰ Generally, the CCD consists of 2^n factorial runs, $2(n)$ axial runs and six center runs, where n is the number of factors.

In recent study, the battery-grade ferrous oxalate was prepared by liquid-phase precipitation method where the variables studied were reaction temperature (A), aging time (B), and ferrous sulfate concentration (C). These three variables together with their respective ranges were chosen based on the results of the literature and preliminary studies. Total number of experiments with different combinations of three variables was 20 for this procedure for each precursor, as calculated from Eq. (1)^{21,22}:

$$N = (2^n + 2n + n_c) = 2^3 + 2(3) + 6 = 20, \quad (1)$$

where N is the total number of experiments required and $n_c (= 6)$ is the number of central points for the system containing three variables.

2.3. Analysis

As for the strict standard of battery-grade ferrous oxalate, two aspects (particle size and purity) were investigated mainly for the performance of ferrous oxalate. Meanwhile, the structural and morphological characterization of ferrous oxalate at different dispersant proportion were also come into notice.

2.3.1. Particle size and purity analysis

The particle size of ferrous oxalate was tested by a laser particle size (LPS) analyzer (LS-POP III, OMEC, China). The result was described in terms of D_{50} , which was regularly used to measure materials' median particle diameter. D_{10} and D_{90} , which represented the grain size diameter occupying 10% and 90% of hole particle size distribution, were regarded as a reference.²³

The purity of ferrous oxalate was obtained by oxidation–reduction titration method. Ferrous oxalate was dissolved by sulfuric solution (containing eight volumes of sulfuric acid and 92 volumes of deionized water) and titrated with potassium permanganate standard solution which had been demarcated accurately.

2.3.2. SEM-EDS, XRD, BET/BJH analysis for ferrous oxalate

The collected samples were washed with deionized water repeatedly, air dried for 8 h at 60°C, and then analyzed by X-ray diffraction (XRD, Rigaku, D/Mac-3c, Japan) to get the crystallographic characterization including crystal structure and cell parameters. SEM with energy dispersive X-ray analysis (SEM-EDS, SU-8020, Hitachi Japan) was done to recognize its crystalline structure, size of the crystal and surface composition. Brunauer–Emmett–Teller (BET/BJH) study was carried out to study the porosity and specific surface area of ferrous oxalate.

3. Result and Discussion

3.1. Factors affecting the synthesis process of ferrous oxalate

The conditions, including reaction temperature, aging time, ferrous sulfate concentration, and dispersant proportion, mainly affect the synthesis process. This section investigates the principle

about the effect of these conditions on the characteristic of ferrous oxalate.

3.1.1. Effect of reaction temperature

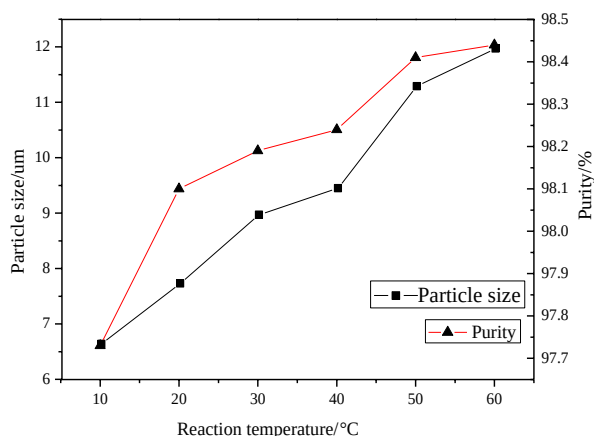
The purity and particle size of ferrous oxalate at different reaction temperature preparing is given in Fig. 1(a). It is indicated that the purity increases from 97.7% to 98.4% as the temperature increase. This effect can be attributed to the reaction and diffusion activation energy for Fe^{2+} and $\text{C}_2\text{O}_4^{2-}$ ions.¹³ The reaction rate and growth rate for grains all accelerate by the increase of temperature, which reduced the reaction and diffusion activation energy. Then, all of these reduce the cladding for the impurity ions and increase the purity of the samples.

Figure 1(a) also revealed that the particle size of samples increases with the increase in the reaction

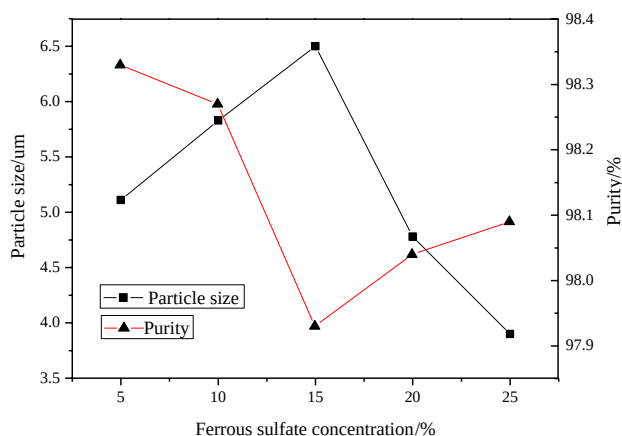
temperature. It can be attributed to the supersaturation of solution and nucleation which can adjust the number and intensity of crystals in the growth process, and also affect the particle size.^{24,25} In addition, considering the increasing of temperature, the movement for particles accelerate, so as to reduce the agglomeration.

3.1.2. Effect of ferrous sulfate concentration

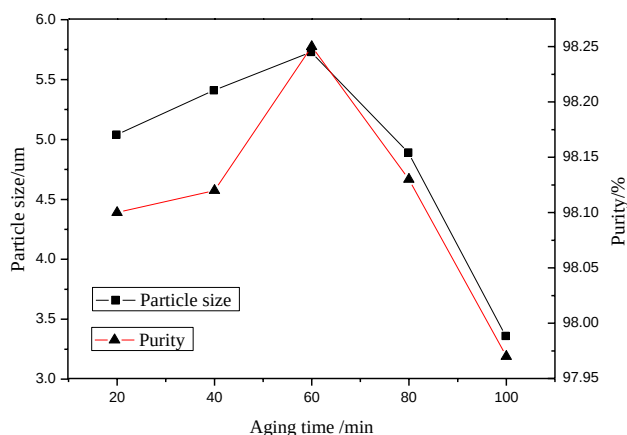
The effect of ferrous sulfate concentration on particle size and purity is shown in Fig. 1(b). It is obvious that particle size increases from $5\mu\text{m}$ to $6.5\mu\text{m}$, and then decreases to $4\mu\text{m}$ with the increasing of concentration. This is because the high concentration of ferrous sulfate can increase the saturation of the reaction system solution and then decrease the growth rate of the crystal. However,



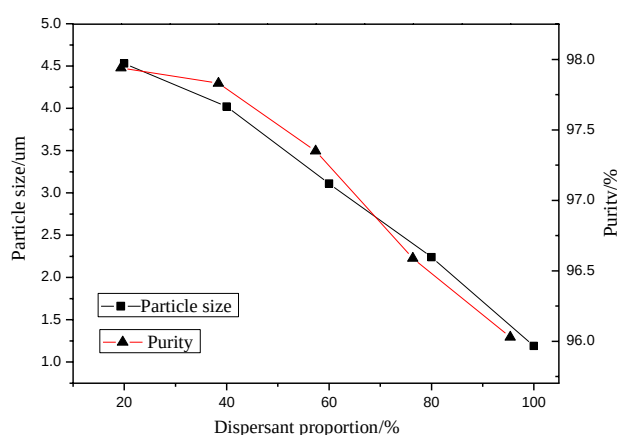
(a)



(b)



(c)



(d)

Fig. 1. Effect of different reaction temperature (a), ferrous sulfate concentration (b), aging time (c), and dispersant proportion (d) on the purity and the particle size.

the effect on purity is opposite. The maximum purity is 98.35% with the concentration of 5%. After this point, the purity of samples are decreased. It is difficult to achieve the reduction of particle size and the improvement of purity only depending on the concentration of ferrous sulfate.

3.1.3. Effect of aging time

The effect of aging time on particle size and purity is shown in Fig. 1(c). The particle size and purity gradually increase before 60 min then go down. It is obvious that the tendency between particle size and purity show same respond with increasing of aging time. Previous reports^{26,27} show that aging time is beneficial to grain growth and second agglomeration of particles, but long aging time can cause the grain to rupture, which can reduce grain size and damage the morphology of grain. On the other hand, shorter time of aging, can result in poor crystallinity and uncontrollable morphology. Hence, the particle of ferrous oxalate grows up because of the increasing of aging time before 60 min, but larger particle are easy to separate into several smaller particles. Also, the smaller particles with bigger specific surface area are easy to adsorb and enrich impurities.

3.1.4. Effect of dispersant proportion

The effects of dispersant proportion were demonstrated in Fig. 1(d). It is obvious that particle size decreased from 4.5 μm to 1.25 μm and purity decreased continuously from 98% to 96% with the increasing of volume ratio for alcohol-to-water. It can be seen that by increasing the dispersant proportion, particle size and purity both decreased by a steady state. The dispersant in solution can reduce particle surface tension, inhibit the sedimentation

and agglomeration between particles, slow the growth rate of crystal, and prevent secondary nucleus.^{28,29}

3.2. Result of response surface optimization study using design expert software

Experiments were planned and conducted by following a five-level and three-factor CCD. According to the experimented results of single factor obtained above the minimum and maximum values were set in software as follows: reaction temperature (*A*) at 23.2°C to 56.8°C, aging time (*B*) between 26.4 min and 93.6 min, concentration of ferrous sulfate (*C*) solution from 1.59% to 18.41% as given in Table 1. Considering the effect of dispersant proportion, it was planned to examine the influence on structural and morphological characterization of ferrous oxalate under optimal conditions, instead of the interaction with other factors. The particle size and purity of ferrous oxalate were taken as responses of the system. The complete design consisted of 20 combinations including six replicates of center point to evaluate the pure error. Based on results of the CCD experimental design shown in Table 2, RSM has been applied to develop the polynomial regression equations and find out the relationship between output response, and input factors. The response surface models of second-order were developed as follow³⁰⁻³²:

$$Y = b_0 + \sum_{i=1}^n b_i x_i + \sum_{i=1}^n b_{ii} x_i^2 + \sum_{i < j}^n b_{ij} x_i x_j + \xi, \quad (2)$$

where *Y* is predicted response, x_i refers to coded variables, b_0, b_i, b_{ii}, b_{ij} are regression coefficients and ξ is statistical error. It was used to estimate the

Table 1. Independent variables and their coded levels for the central composite design.

Numeric factors coded values	Factor values				
	$-\alpha$	-1	0	1	α
A: Reaction temperature (°C)	23.1821	30	40	50	56.8179
B: Aging time (min)	26.3641	40	60	80	93.6359
C: Ferrous sulfate concentration (%)	1.59104	5	10	15	18.409
Constant	Constant level				
Stirring speed (<i>r</i> /min)	$S = 300$				
Oxalic acid concentration (%)	$C_1 = 8$				
Reaction time (min)	$T = 15$				
Dispersant proportion (%)	40%				

Table 2. Central composite design arrangement and results.

Run	Factors						Response	
	Coded variables			Actual variables			Purity/%	Particle size/ μm
	A	B	C	A/ $^{\circ}\text{C}$	B/min	C/%	Observed	
1	-1	-1	-1	30	40	5	99.12	4.49
2	1	-1	-1	50	40	5	98.96	6.89
3	-1	1	-1	30	80	5	99.18	6.19
4	1	1	-1	50	80	5	99.29	7.64
5	-1	-1	1	30	40	15	99.03	9.47
6	1	-1	1	50	40	15	99.23	10.84
7	-1	1	1	30	80	15	98.87	9.32
8	1	1	1	50	80	15	99.31	9.27
9	-1.682	0	0	23.18	60	10	99.05	6.72
10	1.682	0	0	56.82	60	10	99.24	9.43
11	0	-1.682	0	40	26.36	10	99.03	7.85
12	0	1.682	0	40	93.64	10	99.12	7.69
13	0	0	-1.682	40	60	1.59	99.26	5.41
14	0	0	1.682	40	60	18.41	99.18	11.64
15	0	0	0	40	60	10	99.51	9.93
16	0	0	0	40	60	10	99.46	9.85
17	0	0	0	40	60	10	99.49	9.94
18	0	0	0	40	60	10	99.48	9.89
19	0	0	0	40	60	10	99.52	9.73
20	0	0	0	40	60	10	99.47	10.02

response of each set of experimental design and optimized conditions.³³

3.2.1. Model establishment

The corresponding results of CCD experiments were shown in Table 2. The particle size and purity were found to range from 4.49 μm to 11.64 μm and 98.87% to 99.52%, respectively. Run 15–20 at the center point were conducted to determine the experimental error and reproducibility of the data. According to the sequential model sum of squares and model summary statistics for each response, the quadratic model was selected for the response of particle size and purity in software. Through multiple regression analysis on the experimental data, the model for the predicted response could be expressed by following quadratic polynomial equations (in form of coded values):

$$\begin{aligned}
 \text{Purity} = & 96.60868 + 0.067764A + 0.038793B \\
 & + 0.038935C + 0.00031875AB \\
 & + 0.001725AC - 0.0005875BC \\
 & - 0.001285A^2 - 0.0003665B^2 \\
 & - 0.0038133C^2,
 \end{aligned} \tag{3}$$

$$\begin{aligned}
 \text{Particle Size} = & -24.79989 + 0.73803A + 0.33842B \\
 & + 1.30707C - 0.00148125AB \\
 & - 0.006325AC - 0.0052125C \\
 & - 0.00643345A^2 - 0.00187795B^2 \\
 & - 0.01937C^2.
 \end{aligned} \tag{4}$$

As shown in Tables 3 and 4, the regression equations for particle size ($F = 261.25$, $p < 0.0001$) and purity ($F = 181.19$, $p < 0.0001$) were highly significant, demonstrating that the degree of fit was better on border of the independent variables. The quadratic regression models showed a value of determination coefficient (R^2) higher than 0.90 which implied that more than 90% of the variation could be explained by fitted model.¹⁸ The R^2 Adeq is the adequate precision which measures the signal-to-noise, and a ratio greater than 4 is desirable. The proposed model's ratio of 59.283 indicates an adequate signal. Therefore, this model can be used to navigate the design space. In this experiment, the R^2 values for Eqs. (3) and (4) were 0.9936 and 0.9958, respectively. These indicated that 99.36% and 99.58% of the total variation in purity and particle size were attributed to the studied experimental variables. The p -values were used as a tool to check

Table 3. ANOVA for purity response surface quadratic model.

Source	Sum of squares	Mean square	<i>F</i> value	<i>p</i> -value prob > <i>F</i>	
Model	0.75	0.084	181.19	<0.0001	Significant
<i>A</i>	0.061	0.061	131.43	<0.0001	
<i>B</i>	0.016	0.016	33.82	0.0002	
<i>C</i>	0.004379	0.004379	9.50	0.0116	
<i>AB</i>	0.033	0.033	70.54	<0.0001	
<i>AC</i>	0.060	0.060	129.13	<0.0001	
<i>BC</i>	0.028	0.028	59.91	<0.0001	
<i>A</i> ²	0.21	0.21	464.25	<0.0001	
<i>B</i> ²	0.31	0.31	671.99	<0.0001	
<i>C</i> ²	0.13	0.13	284.18	<0.0001	
Residual	0.004609	0.004609			Not significant
Lack of fit	0.001926	0.003851	0.72	0.6377	
Pure error	0.002683	0.005367			
Cor total	0.76				

$R^2 = 0.9939$, $R^2_{\text{Adj}} = 0.9884$, $R^2_{\text{Adeq}} = 41.056$, $\text{CV}(\%) = 0.022$

significance of each coefficient. The coefficient of variance (CV) should not be larger than 10% and is defined as percent ratio between standard error of the estimate and the mean value of the observed responses and this is a measure of reproducibility of the model.³⁴ It can be found that the CV values in both models indicate good reproducibility, with purity 0.022% and particle size 1.98%. In addition, Fig. 2 shows the predicted values and experimental values for purity (*A*) and particle size (*B*). As we see, the predicted values are quite close to the experimental values.

According to the Analysis of variance (ANOVA) for purity response surface quadratic model, it

is obvious that the *p*-value is less than 0.001. Compared with other factors, the effect of reaction temperature (*A*), the interaction between multiple factors (*AB*, *AC*, *BC*), and the quadratic terms of each factors (*A*², *B*², *C*²) appear highly significant. Results also show that the effect of aging time on purity expresses more significant than concentration of ferrous oxalate. It also could be seen that the order of influence on purity for interaction between multiple factors is *AC* ($F = 129.13$) > *AB* ($F = 70.54$) > *BC* ($F = 59.91$).

For particle size response, it could be seen that the effect of two independent factors (*A* and *C*), one interaction condition (*AB*), and three quadratic

Table 4. ANOVA for particle size response surface quadratic model.

Source	Sum of squares	Mean square	<i>F</i> value	<i>p</i> -value prob > <i>F</i>	
Model	68.11	7.57	261.25	<0.0001	Significant
<i>A</i>	6.93	6.93	239.19	<0.0001	
<i>B</i>	0.016	0.016	0.54	0.4805	
<i>C</i>	42.77	42.77	1476.39	<0.0001	
<i>AB</i>	0.70	0.70	24.24	0.0006	
<i>AC</i>	0.80	0.80	27.62	0.0004	
<i>BC</i>	2.17	2.17	75.04	<0.0001	
<i>A</i> ²	5.96	5.96	205.91	<0.0001	
<i>B</i> ²	8.13	8.13	280.72	<0.0001	
<i>C</i> ²	3.38	3.38	116.66	<0.0001	
Residual	0.29	0.029			Not significant
Lack of fit	0.24	0.048	5.02	0.0506	
Pure error	0.048	0.009627			
Cor total	68.40				

$R^2 = 0.9958$, $R^2_{\text{Adj}} = 0.9920$, $R^2_{\text{Adeq}} = 59.283$, $\text{CV}(\%) = 1.98$

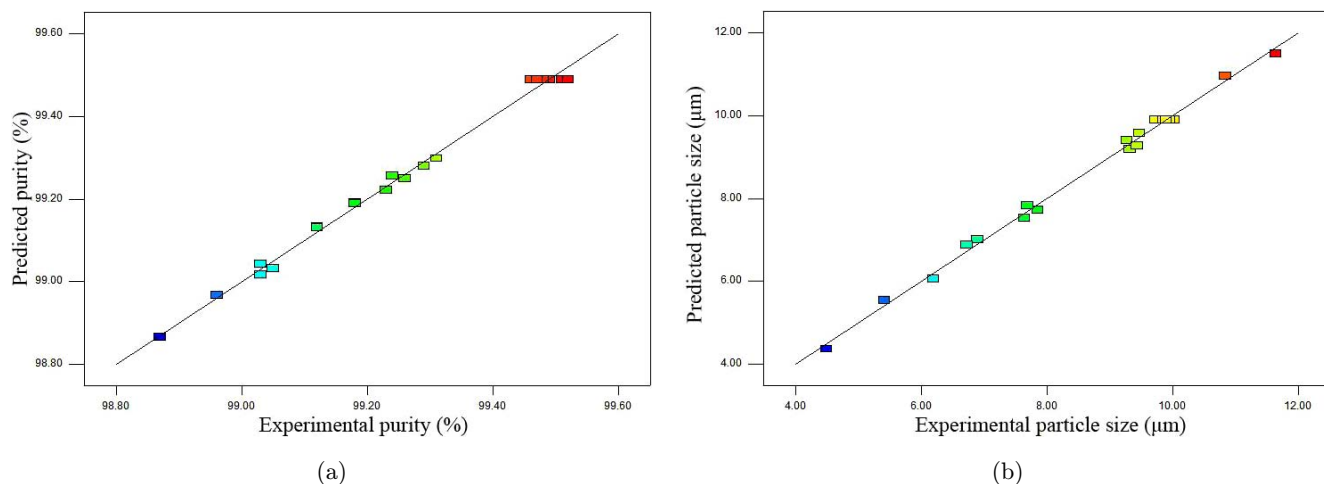


Fig. 2. Distribution of experimentally determined values versus statistically predicted values of (a) purity (%) and (b) particle size (μm).

terms (A^2 , B^2 , C^2) express significant. Compared with the effect to purity, the interaction between multiple factors shows smaller effect on particle.

3.2.2. Response surface analysis

The response surface analysis is another way to predict the relationships among responses and factors and interaction. Figure 3 presents the response surface model in a three-dimensional and contour plots representation reflecting the effects of reaction temperature, aging time, and concentration of ferrous sulfate on purity and particle size.

For the purity of response surface analysis, Fig. 3(a) shows the combined effects of reaction

temperature and aging time on the purity. These plots present the response in function of two factors, keeping the other variable constant at its middle level (center value of the testing ranges). The tortuose surface and oval contour plot shows a strong interaction between these two factors.³⁵ It can be seen that the effect of reaction temperature on purity is more significant than aging time, when their values are both in center level. Higher purity of ferrous oxalate is obtained at reaction temperature between 43 and 45.5°C and aging time between 61 min and 65 min. This demonstrates that the effects of reaction temperature (A) and aging time (B) on purity are significant. It agrees well with the results in Table 3. When purity remains to be

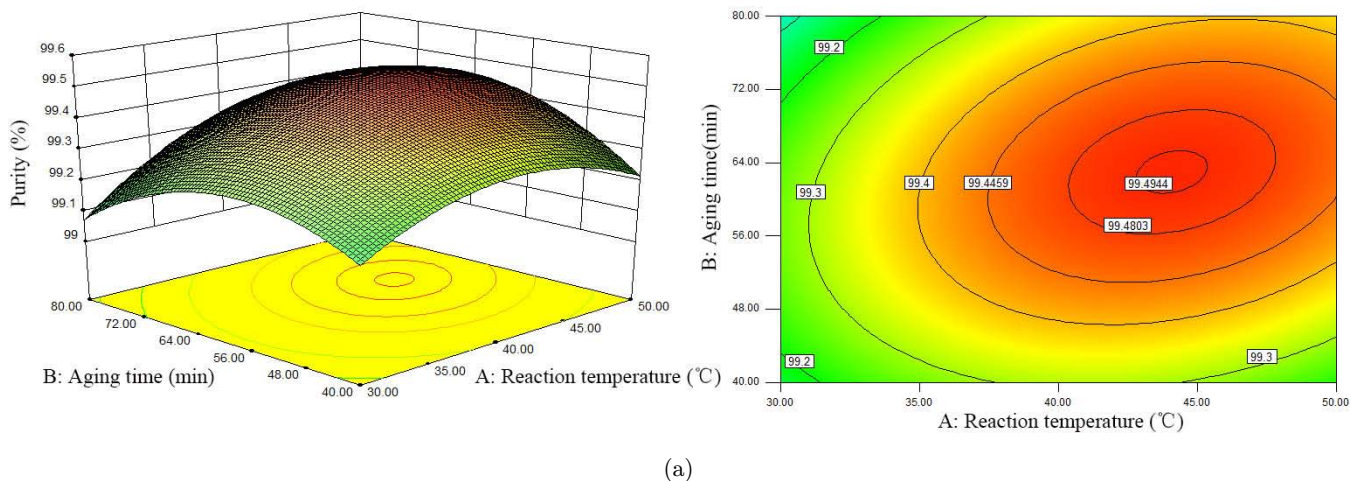
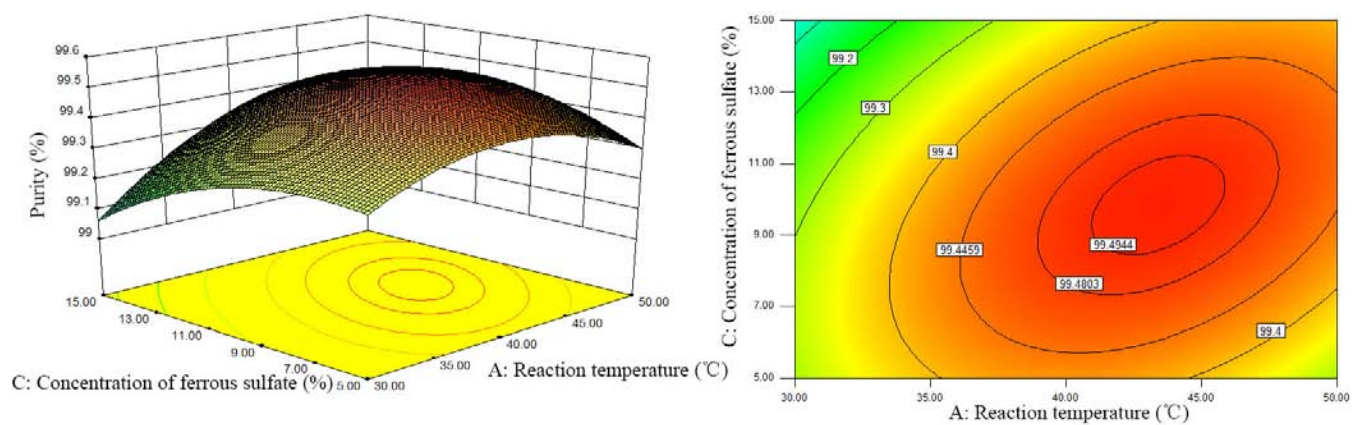
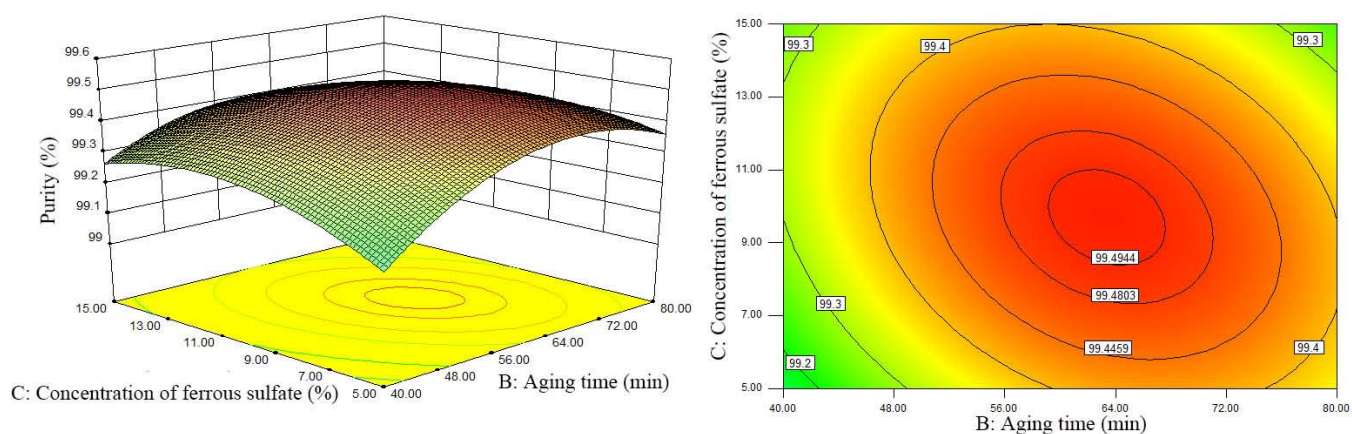


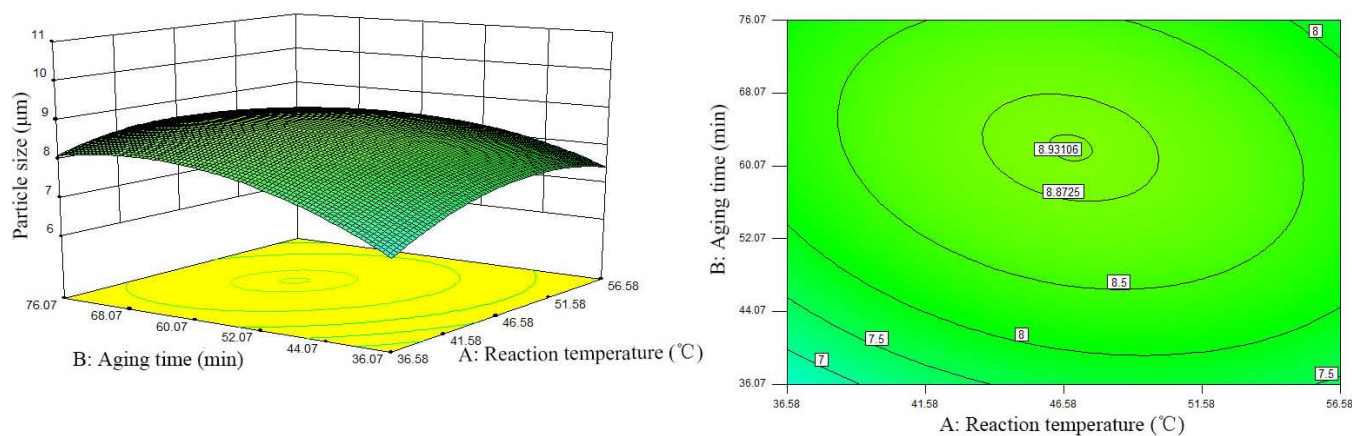
Fig. 3. Response surface (3D) showing the effects of different extraction parameters on the purity (a)–(c) and particle size (e)–(f). ((a) Ferrous sulfate concentration is 11.35%; (b) aging time is 65.95 min; (c) reaction temperature is 41.89°C); (d) ferrous sulfate concentration is 6.89%; (e) aging time is 40.01 min and (f) reaction temperature is 40.01°C).



(b)



(c)



(d)

Fig. 3. (Continued)

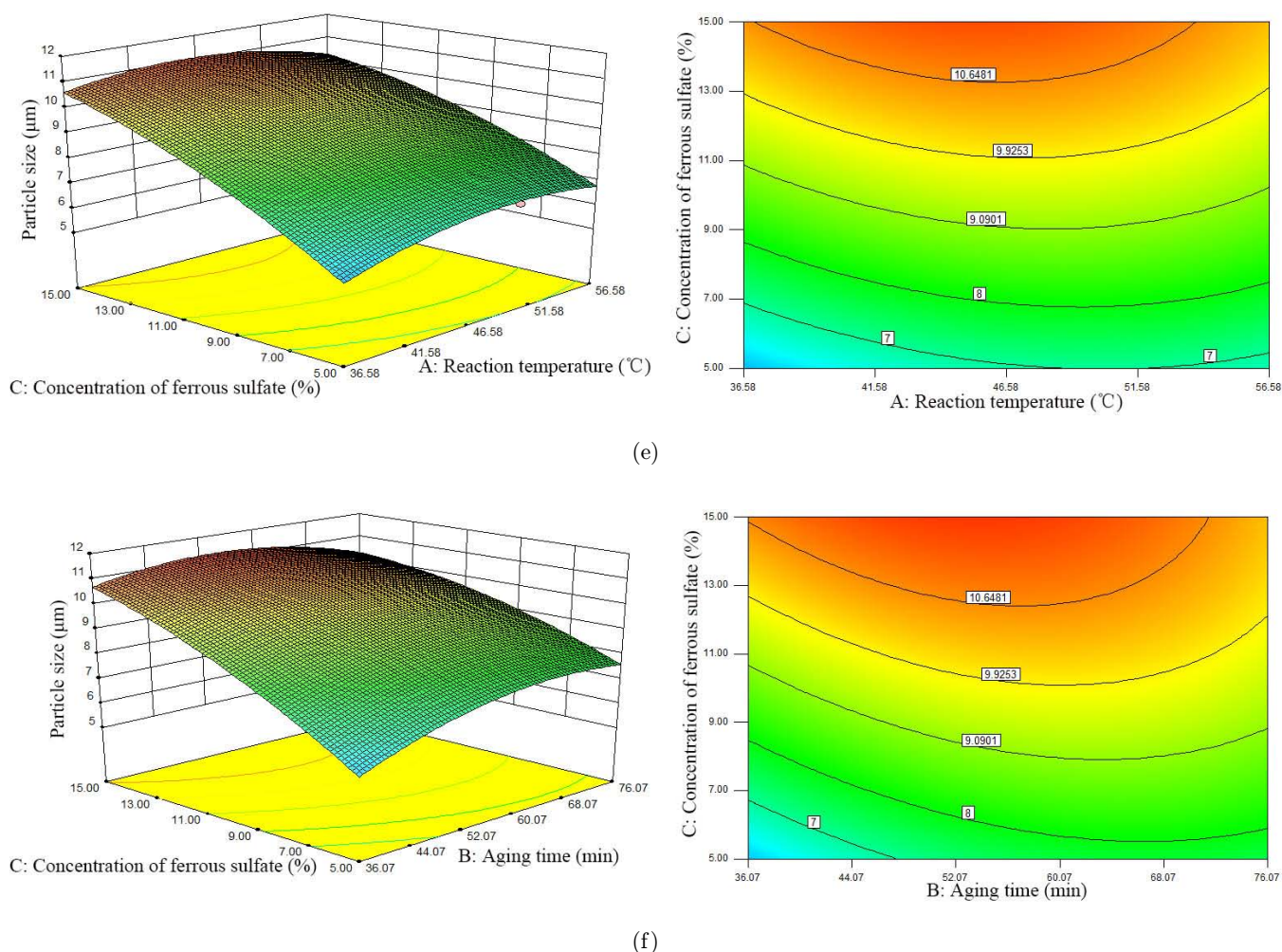


Fig. 3. (Continued)

99.45%, initially with the increasing of reaction temperature, the aging time also increases. And when reaction temperature is above 44°C , aging time begins to decrease. This explains that aging time and reaction temperature have significant interaction. In order to achieve the same corresponding index value, we can control both of them within a scope of certain level. The effects of reaction temperature (A) and ferrous sulfate concentration (C) on purity are shown in Fig. 3(b). It can be seen that the purity increases with the concentration of ferrous sulfate amount increasing. In the contour diagram, the smallest ellipse and deepest color represent the maximum predicted value.³⁶ So, it is easily discovered that purity of ferrous oxalate can achieve higher, when reaction temperature and ferrous sulfate concentration are range from 41°C to 45.5°C , and from 8.5% to 11.2%, respectively. Figure 3(c) indicates that the aging time (B) and

concentration of ferrous sulfate (C) also have more significant on the purity. When both the factors are at the threshold levels of 60–68 min and 8.5–11%, the purity can reach maximum. Under the same conditions of aging time or concentration of ferrous sulfate, the purity of ferrous oxalate first increases and then decreases with the increase of the concentration of ferrous sulfate or aging time.

For the particle size of response surface analysis, the effects of reaction temperature (A) and aging time (B) on particle size are shown in Fig. 3(d). The particle size increases when the reaction temperature extended from 36.58° to 46.58°C , and then decreases. Similarly, the particle size is also found to increase with increase of the aging time from 36.07 min to 65.07 min, and then decreases. So, the lower reaction temperature and shorter aging time are helpful to prepare ferrous oxalate with smaller particle size and higher purity. Figure 3(e) shows

the relationship between reaction temperature (A) and aging time (B) on particle size. It is observed that effects of particle size of ferrous oxalate depends on the concentration of ferrous sulfate. Lower concentration of ferrous sulfate leads to smaller particle size. When the concentration of ferrous sulfate is lower than 11%, the particle size of material is obviously decreased. As shown in Fig. 3(f), it illustrates the effect of aging time and concentration of ferrous sulfate on the particle size, with reaction temperature at zero level (40.01°C). Compared with the concentration of ferrous sulfate, the effect of aging time on particle size is small, and its response surface analysis result is consistent with the result of single factor analysis, which shows that the time of aging has a little relationship and interaction with other variables.

As shown in these figures, the high purity of ferrous oxalate can be achieved, when the reaction temperature and aging time are range from 41°C to 45.5°C , and from 61 min to 65 min, and the ferrous sulfate concentration is from 8.5% to 11.2%. In addition, the reaction temperature was the main factor effect on the purity and particle size among the three factors.

3.2.3. Optimization of preparation parameters

The main purpose of this research is to determine the optimal process parameters in order to prepare ferrous oxalate products with high purity and small particle size. However, it is difficult to get a unified development for both these responses improvement under the same condition because the interest region of factors is different. So, the CCD was used by Design-Expert software in order to compromise between these two responses.

The predicted and experimental results of purity and particle size obtained at optimum conditions are listed in Table 5. The optimum purity is obtained by using reaction temperature, aging time and ferrous sulfate concentration of 43.19°C ,

63.73 min and 9.96%, respectively. And compared with the experimental value (99.85%), it is close to the maximum response of model predicted (99.5022%). Considering both of the responses comprehensively, the optimum options are as follows: reaction temperature 31.32°C , aging time 56.52 min, and concentration of ferrous sulfate 5%. It is observed that the experimental values (99.69% and $4.92\ \mu\text{m}$) obtained keep in good agreement with the values predicted (99.3254% and $6.099\ \mu\text{m}$) from the models.

3.3. Structural and morphological characterization of ferrous oxalate prepared at different dispersant proportion under optimum condition

3.3.1. Structural analysis

According to early literature, $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ crystallizes in two allotropic forms, α monoclinic (space group $C2/c$) and β orthorhombic (space group $Cccm$).^{8,37} As the XRD patterns shown in Fig. 4(a), all of the diffraction peaks for the ferrous oxalate samples are readily indexed to the orthorhombic β -phase (ICDD PDF no. 022-0635) with $a = 12.26\ \text{\AA}$, $b = 5.57\ \text{\AA}$ and $c = 15.48\ \text{\AA}$. XRD patterns show that all the samples have narrow peak width which indicates larger size and higher crystallinity degrees. There are no impurities in all samples. Figure 4(b) shows the EDS spectrum of two selected samples (dispersant proportion 40% and 100%). The EDS patterns includes the spectrum peaks of C, Fe and O, the results agree with XRD analysis.

3.3.2. Surface morphology

Figure 5 shows the SEM micrographs of the samples at the different dispersant proportions. The images show special morphologies and different sizes of samples. Ferrous oxalate prepared in low proportion

Table 5. Optimization runs for RSM model.

Number	$A/^{\circ}\text{C}$	B/min	$C/\%$	Desirability	Purity/%		Particle size/ μm	
					Theoretical value	Experimental value	Theoretical value	Experimental value
1	43.19	63.73	9.96	0.973 Selected	99.5022	99.85	—	—
2	30.00	40.00	5.00	1.00 Selected	—	—	4.3652	7.16
3	31.32	56.52	5.00	0.737 Selected	99.3254	99.69	6.099	4.92

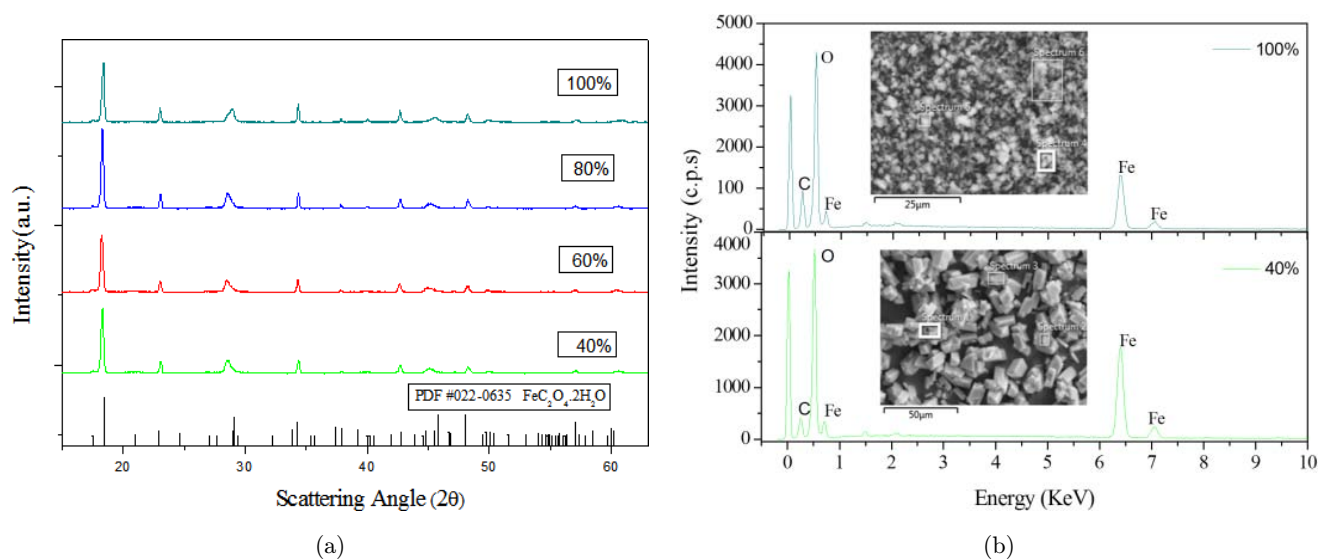


Fig. 4. XRD patterns (a) and EDS spectrum (b) for the ferrous oxalate with different dispersant proportion.

of dispersant exhibits micrometer-sized and rod-shaped-particles, with 3–5 μm width, 15–20 μm length, and smooth surface. It easily observed the cross-section under high magnification. The individual particle grows from smaller particles and then forms into rod-shaped structure. It can be considered that the dispersant can reduce the surface tension of single rod particle, inhibit the sedimentation and agglomeration between particles.²⁸

On the other hand, rectangular rods in the cross-section are formed in high proportion of dispersant. The single rods have lengths in range of 1.5–2.5 μm and widths with 0.5–1 μm in range of submicrometer to micrometer.

As shown in Fig. 6, the reaction route for morphological variation can be observed. Fe ion and oxalic acid do not react immediately because of large differences in their ionization abilities in an

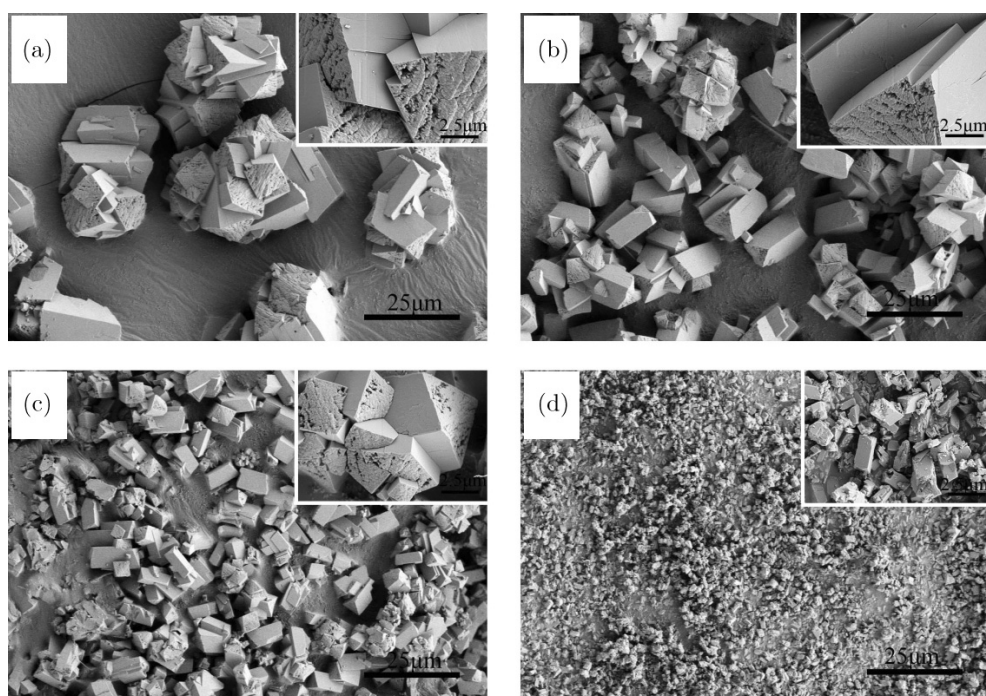


Fig. 5. SEM micrograph and magnified view (a) 40%; (b) 60%; (c) 80% and (d) 100%.

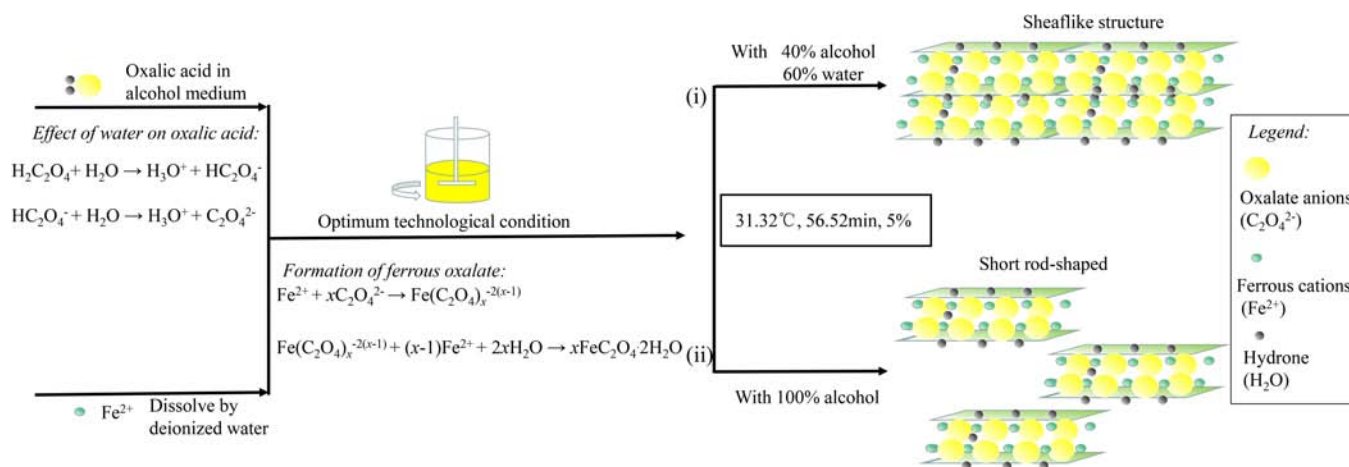


Fig. 6. Reaction route for morphological variation.

organic solvent medium.¹⁴ The oxalate complex $\text{Fe}(\text{C}_2\text{O}_4)_x^{-2(x-1)}$ is the dominating species at a special pH value and solution composition,³⁸ which can slow down the growth rate of grain and then make sample particle more smaller. Generally, the anisotropic growth of crystal structure is a result of large differences in the surface energy between crystallographic planes.³⁹ This means that the effect of dispersant is the key factor in morphological variations. Hence, different morphologies of ferrous oxalate can be achieved by controlling the grain growth using different dispersant proportion.

3.3.3. BET/BJH surface area and pore volume

To study the porosity and specific surface area of ferrous oxalate, BET gas sorption was carried out.

The nitrogen adsorption-desorption isotherms (at 77 K) of these samples are shown in Fig. 7. As we see, the initial gases absorbed for each sample (40%, 60%, 80% and 100%) are 6.1, 6.5, 2.4 and 3.1 cm³/g⁻¹, respectively. The sharp decline in the desorption curves indicates that the samples have mesoporous structures with typical Langmuir IV type curves.¹⁴ The BET surface areas were found to be 44.644, 44.041, 41.75 and 31.681 m²/g, respectively. It is also observed that a sudden increase appears due to aggregation of the particles when the relative pressure P/P_0 is greater than 0.85. The plot of pore size distribution determined in Fig. 8 shows that there is a same dominant peak around 3.5 nm for the samples. The relatively large specific surface area and high porosity offer large materials/electrolyte contact area and promote Li⁺ diffusion in Li-ion battery.⁴⁰

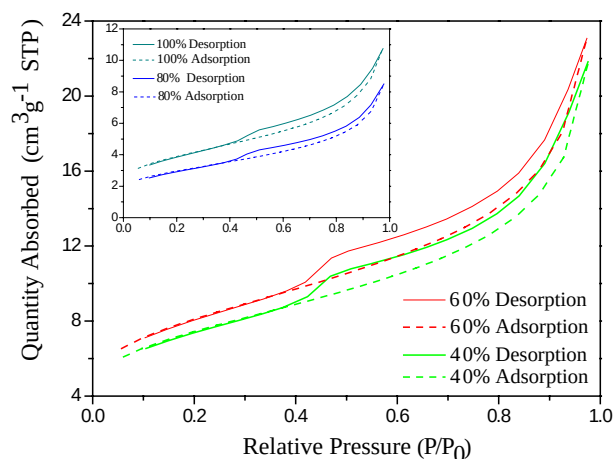


Fig. 7. Nitrogen absorption-desorption isotherm curves.

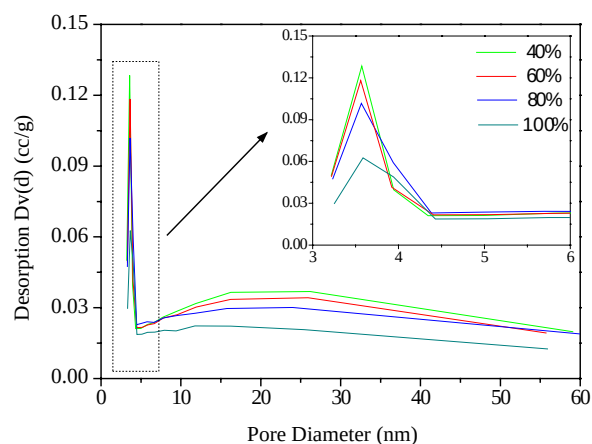


Fig. 8. BJH pore-size distribution.

4. Conclusion

RSM was successfully used to investigate the effects of reaction conditions to reduce particle size and improve purity of ferrous oxalate. The optimum ferrous oxalate preparation conditions are obtained by using 31.32°C reaction temperature, 56.52 aging time, and 5% concentration of ferrous sulfate, resulting in 99.69% of purity and 4.92 μm of particle size. Through the analysis of the response surface, reaction conditions were found to have significant effects on ferrous oxalate. The dispersant could effectively change the surface energy between crystallographic planes, then result in anisotropic growth of the crystal structure and change the morphology of synthetic products. Ferrous oxalate prepared at this condition was demonstrated high purity, small particle size, regular morphology and good porosity.

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